

EXPERIMENT – 34

Implementation of ECB, CBC, and CFB Modes of Operation with Padding

CODE:

```
#include <stdio.h>

void print(int a[], int n)
{
    for(int i = 0; i < n; i++)
        printf("%d", a[i]);
}

/* Simple XOR cipher for demonstration */
void encrypt(int in[], int out[], int key[])
{
    for(int i = 0; i < 8; i++)
        out[i] = in[i] ^ key[i];
}

int main()
{
    int P[2][8] = {
        {1,0,1,0,1,0,1,0},
        {1,1,0,0,1,1,0,0}
    };

    int key[8] = {1,1,0,0,1,0,1,0};
    int IV[8] = {0,1,0,1,0,1,0,1};
```

```
int E[2][8], C[2][8], F[2][8], temp[8], stream[8];
```

```
/* ECB */
```

```
for(int i = 0; i < 2; i++)  
    encrypt(P[i], E[i], key);
```

```
/* CBC */
```

```
for(int j = 0; j < 8; j++)  
    temp[j] = P[0][j] ^ IV[j];
```

```
encrypt(temp, C[0], key);
```

```
for(int j = 0; j < 8; j++)  
    temp[j] = P[1][j] ^ C[0][j];
```

```
encrypt(temp, C[1], key);
```

```
/* CFB */
```

```
encrypt(IV, stream, key);  
for(int j = 0; j < 8; j++)  
    F[0][j] = P[0][j] ^ stream[j];
```

```
encrypt(F[0], stream, key);  
for(int j = 0; j < 8; j++)  
    F[1][j] = P[1][j] ^ stream[j];
```

```
printf("ECB: ");
```

```
print(E[0], 8); printf(" ");
```

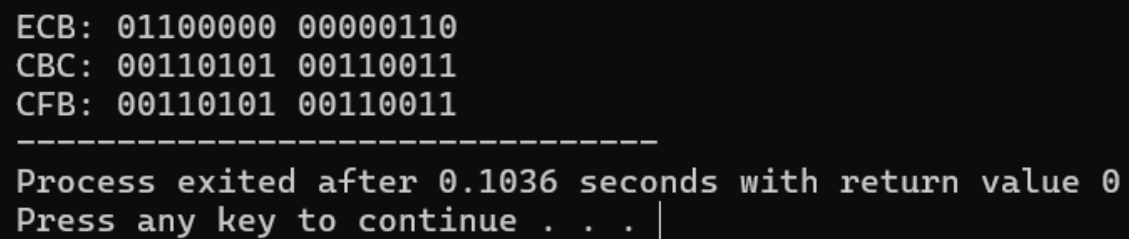
```
print(E[1], 8);

printf("\nCBC: ");
print(C[0], 8); printf(" ");
print(C[1], 8);

printf("\nCFB: ");
print(F[0], 8); printf(" ");
print(F[1], 8);

return 0;
}
```

OUTPUT:



```
ECB: 01100000 00000110
CBC: 00110101 00110011
CFB: 00110101 00110011
-----
Process exited after 0.1036 seconds with return value 0
Press any key to continue . . . |
```